London Cycles: Five Stories, One Fleet

Interactive Dashboard Proposal · M2 Project · May 2026

Overview

A single self-contained **index.html** dashboard — no server, no framework — that presents five data-driven insights through scrollytelling chapters. Each chapter combines a narrative sentence with one interactive chart or map. Hosted on GitHub Pages; shareable as a file.

FORMAT	CHARTS	MAPS	HOSTING
Self-contained HTML	Plotly.js	Leaflet.js	GitHub Pages

Five Chapters

01 Two Markets, One Fleet

Animated · Toggle

Plotly animated heatmap with a **Weekday / Weekend toggle**. A play button animates through hours 0–23, lighting up demand row by row. The bimodal weekday shape vs. flat weekend pattern is revealed in seconds.

Why: The contrast is the headline finding — animation makes it viscerally obvious without reading a caption.

02 The Redistribution Problem

Map · Filter

Leaflet map with station markers: **red = source, blue = sink, grey = balanced**. Marker size scales with net outflow. A "Show top 15 imbalanced" button dims all other stations. Clicking a marker shows imbalance %, departures vs arrivals.

Why: Station imbalance is a spatial problem — a map makes it immediately clear where vans need to go.

03 Eight Years of Seasons

Range slider · Anomaly flags

Plotly line chart (2015–2023) with a **range slider** to zoom any period. COVID spike, Queen's death dip, and disrupted months appear as **annotated markers** directly on the line — hovering reveals the event label.

Why: The range slider lets users explore any era; anomaly markers surface surprise events without burying them in a table.

04 The Third Market: Nightlife

Scroll-triggered Map

Two panels: a Plotly bar chart (midnight rentals by day, Saturday highlighted purple, bars grow on scroll) + a Leaflet mini-map with **pulsing dots** at the top 15 nightlife stations, sized by volume. Tooltip shows station name and avg trip duration.

Why: The map directly answers the operational question — where to pre-position bikes on Friday night.

05 The Risk Register

Search Sortable table

Plotly scatter (x = total rentals, y = hours ridden, colour = risk category). A **bike ID search box** highlights any bike instantly. Below: a sortable top-20 risk table with colour-coded rows — red for high-risk bikes with no maintenance record.

Why: Turns an abstract fleet audit into a live risk register — actionable and scannable.

Implementation Plan

Step	Detail
1. Data export	Python script queries all 5 mart tables from BigQuery and writes `data.js` with embedded JSON
2. HTML shell	Sticky nav, five `<section>` chapters, scroll-triggered animations via IntersectionObserver
3. Charts	Plotly.js for all non-map charts (heatmap, line, bar, scatter)
4. Maps	Leaflet.js using lat/long from `dim_station` and `mart_nightlife_demand`
5. Hosting	Push `index.html` + `data.js` to GitHub Pages — zero infrastructure